MATH 342W / 642 / RM 742 Spring 2026 HW #3

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Problem 1

These are questions about Silver's book, chapters 3-6. For all parts in this question, answer using notation from class (i.e. $t, f, g, h^*, \delta, \epsilon, e, t, z_1, \ldots, z_t, \mathbb{D}, \mathcal{H}, \mathcal{A}, \mathcal{X}, \mathcal{Y}, X, y, n, p, x_{\cdot 1}, \ldots, x_{\cdot p}$, $x_{1\cdot}, \ldots, x_{n\cdot}$, etc.

(a) [difficult] Chapter 4 is all about predicting weather. Broadly speaking, what is the problem with weather predictions? Make sure you use the framework and notation from class. This is not an easy question.

> The main difficulty is that the atmosphere is a dynamic and chaotic system, so small measurement errors in X can produce large differences in the output. Thus the main problem is ignorance error $\delta = f - t$, caused by imperfect observation of the true atmospheric state.

(b) [easy] Why does the weatherman lie about the chance of rain? And where should you go if you want honest forecasts?

> People dislike unexpected rain more than a forecast that slightly overpredicts rain.

(c) [difficult] Chapter 5 is all about predicting earthquakes. Broadly speaking, what is the problem with earthquake predictions? It is *not* the same as the problem of predicting weather. Read page 162 a few times. Make sure you use the framework and notation from class.

> The problem is that earthquakes have very weak predictive signals, so even good models cannot predict exact events. The main difficulty is large irreducible noise ϵ rather than measurement error.

(d) [easy] Silver has quite a whimsical explanation of overfitting on page 163 but it is really educational! What is the nonsense predictor in the model he describes?

The nonsense predictor is memorizing the specific combinations of the three locks based on their color (red, black, blue).

(e) [easy] John von Neumann was credited with saying that "with four parameters I can fit an elephant and with five I can make him wiggle his trunk". What did he mean by that and what is the message to you, the budding data scientist?

Von Neumann meant that with enough parameters a model can fit almost any dataset. This produces low training error but poor prediction, showing the danger of overfitting.

(f) [difficult] Chapter 6 is all about predicting unemployment, an index of macroeconomic performance of a country. Broadly speaking, what is the problem with unemployment predictions? It is *not* the same as the problem of predicting weather or earthquakes. Make sure you use the framework and notation from class.

The problem is that the economy changes over time because people respond to forecasts and policies, so the relationship learned from past data D may not hold in the future. This leads to model misspecification.

(g) [E.C.] Many times in this chapter Silver says something on the order of "you need to have theories about how things function in order to make good predictions." Do you agree? Discuss.

Problem 2

These are questions about multivariate linear model fitting using the least squares algorithm.

(a) [difficult] Derive $\frac{\partial}{\partial \boldsymbol{c}}\left[\boldsymbol{c}^\top A \boldsymbol{c}\right]$ where $\boldsymbol{c} \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$ but *not* symmetric. Get as far as you can.

Let $f(c) = c^\top A c$ where $c \in \mathbb{R}^n$ and $A \in \mathbb{R}^{n\times n}$.
Write $f(c) = \sum_{i=1}^n \sum_{j=1}^n c_i a_{ij} c_j$.
Take the derivative with respect to c_k.
c_k appears twice in the sum, so
$\frac{\partial f}{\partial c_k} = \sum_{j=1}^n a_{kj} c_j + \sum_{i=1}^n a_{ik} c_i$.
The first term is the kth element of Ac and the second term is the kth element of $A^\top c$.
So
$\frac{\partial}{\partial c}(c^\top A c) = Ac + A^\top c = (A + A^\top)c$.
(help from ChatGPT)

(b) [easy] Given matrix $X \in \mathbb{R}^{n \times (p+1)}$, full rank and first column consisting of the $\mathbf{1}_n$ vector, rederive the least squares solution $\boldsymbol{b}$ (the vector of coefficients in the linear model shipped in the prediction function g). No need to rederive the facts about vector derivatives.

OLS chooses b to minimize
$SSE(b) = ||y - Xb||^2 = (y - Xb)^\top (y - Xb)$.
Expand:
$SSE(b) = y^\top y - 2b^\top X^\top y + b^\top X^\top Xb$.
Take the derivative with respect to b:
$\frac{\partial}{\partial b} SSE(b) = -2X^\top y + 2X^\top Xb$.
Set this equal to 0:
$-2X^\top y + 2X^\top Xb = 0$.
So
$X^\top Xb = X^\top y$.
Since X has full rank, $X^\top X$ is invertible, so
$b = (X^\top X)^{-1} X^\top y$.

(c) [harder] Consider the case where $p = 1$. Show that the solution for $\boldsymbol{b}$ you just derived in (b) is the same solution that we proved for simple regression. That is, the first element of $\boldsymbol{b}$ is the same as $b_0 = \bar{y} - r\frac{s_y}{s_x}\bar{x}$ and the second element of $\boldsymbol{b}$ is $b_1 = r\frac{s_y}{s_x}$.

When $p = 1$, the normal equations are
$nb_0 + b_1 \sum_{i=1}^n x_i = \sum_{i=1}^n y_i$ and $b_0 \sum_{i=1}^n x_i + b_1 \sum_{i=1}^n x_i^2 = \sum_{i=1}^n x_i y_i$.
From the first equation,
$b_0 = \bar{y} - b_1 \bar{x}$.
Substitute this into the second equation and simplify:
$b_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$.
Since $r = \frac{s_{xy}}{s_x s_y}$, $\rightarrow b_1 = \frac{s_{xy}}{s_x^2} = r\frac{s_y}{s_x}$.
Then $b_0 = \bar{y} - b_1 \bar{x} = \bar{y} - r\frac{s_y}{s_x}\bar{x}$.
So the OLS solution is the same as the simple regression solution: $b_0 = \bar{y} - r\frac{s_y}{s_x}\bar{x}$ and $b_1 = r\frac{s_y}{s_x}$.

(d) [easy] If X is rank deficient, how can you solve for $\boldsymbol{b}$? Explain in English.

If X is rank deficient, then $X^\top X$ is not invertible, so we cannot compute
$b = (X^\top X)^{-1} X^\top y$.
This happens when some columns of X are linear combinations of others.
To solve the problem, we remove the redundant columns of X so that X has full rank, and then compute
$b = (X^\top X)^{-1} X^\top y$.

(e) [difficult] Prove $\text{rank}\,[X] = \text{rank}\,\left[X^\top X\right]$.

First show $null(A) \subseteq null(A^\top A)$.
Let $v \in null(A)$. Then $Av = 0$.
Multiply by $A^\top$: $A^\top Av = A^\top(0) = 0$.
So $v \in null(A^\top A)$.
We want to show $null(A^\top A) \subseteq null(A)$.
Let $v \in null(A^\top A)$.
Then $A^\top Av = 0$.
Multiply on the left by $v^\top$:
$v^\top A^\top Av = (Av)^\top(Av) = ||Av||^2$.
So $||Av||^2 = 0$, so $Av = 0$.
Therefore $v \in null(A)$.
Hence $null(A) = null(A^\top A)$.
By the rank-nullity theorem, this implies
$rank(A) = rank(A^\top A)$.

(f) [harder] [MA] If $p = 1$, prove $r^2 = R^2$ i.e. the linear correlation is the same as proportion of sample variance explained in a least squares linear model.

When $p = 1$, we know from simple regression that
$b_1 = r\frac{s_y}{s_x}$.
Also, $\hat{y}_i = b_0 + b_1 x_i$
and since $b_0 = \bar{y} - b_1\bar{x}$, we get $\hat{y}_i - \bar{y} = b_1(x_i - \bar{x})$.
So $SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 = \sum_{i=1}^n b_1^2(x_i - \bar{x})^2 = b_1^2 \sum_{i=1}^n (x_i - \bar{x})^2$.
Substitute $b_1 = r\frac{s_y}{s_x}$:
$SSR = \left(r\frac{s_y}{s_x}\right)^2 \sum_{i=1}^n (x_i - \bar{x})^2$.
Since $\sum_{i=1}^n (x_i - \bar{x})^2 = (n-1)s_x^2$ and $SST = \sum_{i=1}^n (y_i - \bar{y})^2 = (n-1)s_y^2$,
we have $SSR = r^2\frac{s_y^2}{s_x^2}(n-1)s_x^2 = r^2(n-1)s_y^2 = r^2 SST$.
Therefore $R^2 = \frac{SSR}{SST} = r^2$.

(g) [harder] Prove that $g([1\ \bar{x}_1\ \bar{x}_2\ \ldots\ \bar{x}_p]) = \bar{y}$ in OLS.

In OLS, $g([1, x_1, \ldots, x_p]) = b_0 + b_1 x_1 + \cdots + b_p x_p$.
So $g([1, \bar{x}_1, \ldots, \bar{x}_p]) = b_0 + b_1 \bar{x}_1 + \cdots + b_p \bar{x}_p$.
From the normal equations, $X^\top X b = X^\top y$.
Looking at the first row, since the first column of X is 1_n, we get $\sum_{i=1}^n y_i = \sum_{i=1}^n \hat{y}_i$.
But $\hat{y}_i = b_0 + b_1 x_{i1} + \cdots + b_p x_{ip}$, so $\sum_{i=1}^n y_i = \sum_{i=1}^n (b_0 + b_1 x_{i1} + \cdots + b_p x_{ip})$.
This becomes $\sum_{i=1}^n y_i = n b_0 + b_1 \sum_{i=1}^n x_{i1} + \cdots + b_p \sum_{i=1}^n x_{ip}$.
Divide by n: $\bar{y} = b_0 + b_1 \bar{x}_1 + \cdots + b_p \bar{x}_p$.
Therefore $g([1, \bar{x}_1, \ldots, \bar{x}_p]) = \bar{y}$.

(h) [harder] Prove that $\bar{e} = 0$ in OLS.

In OLS, the residuals are $e_i = y_i - \hat{y}_i$.
So $\bar{e} = \frac{1}{n} \sum_{i=1}^n e_i = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)$.
From the normal equations, $X^\top X b = X^\top y$.
Looking at the first row, since the first column of X is 1_n, we get $\sum_{i=1}^n y_i = \sum_{i=1}^n \hat{y}_i$.
So $\sum_{i=1}^n (y_i - \hat{y}_i) = 0$.
So $\sum_{i=1}^n e_i = 0$.
Divide by n: $\bar{e} = \frac{1}{n} \sum_{i=1}^n e_i = 0$.
Therefore $\bar{e} = 0$.

(i) [difficult] If you model $\boldsymbol{y}$ with one categorical nominal variable that has levels A, B, C, prove that the OLS estimates look like $\bar{y}_A$ if $x = A$, $\bar{y}_B$ if $x = B$ and $\bar{y}_C$ if $x = C$. You can choose to use an intercept or not. Likely without is easier.

Without an intercept, define indicator variables: $x_{iA} = 1$ if $x_i = A$, $x_{iB} = 1$ if $x_i = B$, and $x_{iC} = 1$ if $x_i = C$.
The model is $g(x_i) = b_A x_{iA} + b_B x_{iB} + b_C x_{iC}$.
So if $x_i = A$, $g(x_i) = b_A$. If $x_i = B$, $g(x_i) = b_B$. If $x_i = C$, $g(x_i) = b_C$.
The sum of squared errors is $SSE = \sum_{x_i=A} (y_i - b_A)^2 + \sum_{x_i=B} (y_i - b_B)^2 + \sum_{x_i=C} (y_i - b_C)^2$.
Minimizing SSE gives $\sum_{x_i=A} (y_i - b_A) = 0$, $\sum_{x_i=B} (y_i - b_B) = 0$, and $\sum_{x_i=C} (y_i - b_C) = 0$.
So $b_A = \bar{y}_A$, $b_B = \bar{y}_B$, and $b_C = \bar{y}_C$.
Therefore $g(x) = \bar{y}_A$ if $x = A$, $g(x) = \bar{y}_B$ if $x = B$, and $g(x) = \bar{y}_C$ if $x = C$.
(Help from ChatGPT)

(j) [harder] [MA] Prove that the OLS model always has $R^2 \in [0, 1]$.

Problem 3

These are questions related to the concept of orthogonal projection, QR decomposition and its relationship with least squares linear modeling.

(a) [easy] Let H be the orthogonal projection onto $\text{colsp}\,[\boldsymbol{X}]$ where $\boldsymbol{X}$ is an $n \times (p+1)$ matrix with all columns linearly independent. What is $\text{rank}(H)$?

Since H is the orthogonal projection onto $\text{colsp}\,[\boldsymbol{X}]$, we know $\text{colsp}\,[H] = \text{colsp}\,[\boldsymbol{X}]$.
Because the columns of $\boldsymbol{X}$ are linearly independent, $\text{rank}(X) = p+1$.
Therefore $\text{rank}(H) = p+1$.
(Had to change formatting to fix compiling glitch)

(b) [easy] Simplify HX by substituting for H.

Substitute $H = \boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top$.
Then $HX = \boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{X}$.
Since $(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top \boldsymbol{X} = \boldsymbol{I}_{p+1}$, this becomes $H\boldsymbol{X} = \boldsymbol{X}\boldsymbol{I}_{p+1} = \boldsymbol{X}$.
(Had to change formatting to fix compiling glitch)

(c) [harder] What does your answer from the previous question mean conceptually?

Conceptually, $H\boldsymbol{X} = \boldsymbol{X}$ means that every column of $\boldsymbol{X}$ is unchanged by the projection, because each column of $\boldsymbol{X}$ already lies in $\text{colsp}\,[\boldsymbol{X}]$.

(d) [difficult] Let $\boldsymbol{X}'$ be the matrix of $\boldsymbol{X}$ whose columns are in reverse order meaning that $\boldsymbol{X} = [\mathbf{1}_n \vdots \boldsymbol{x}_{\cdot 1} \vdots \ldots \vdots \boldsymbol{x}_{\cdot p}]$ and $\boldsymbol{X}' = [\boldsymbol{x}_{\cdot p} \vdots \ldots \vdots \boldsymbol{x}_{\cdot 1} \vdots \mathbf{1}_n]$. Show that the projection matrix that projects onto $\text{colsp}\,[X]$ is the same exact projection matrix that projects onto $\text{colsp}\,[X']$.

Reversing the order of the columns does not change the subspace they span.
So $\text{colsp}\,[\boldsymbol{X}'] = \text{colsp}\,[\boldsymbol{X}]$.
The orthogonal projection matrix onto $\text{colsp}\,[\boldsymbol{X}]$ is $H = \boldsymbol{X}(\boldsymbol{X}^\top \boldsymbol{X})^{-1}\boldsymbol{X}^\top$.
The orthogonal projection matrix onto $\text{colsp}\,[\boldsymbol{X}']$ is $H' = \boldsymbol{X}'(\boldsymbol{X}'^\top \boldsymbol{X}')^{-1}\boldsymbol{X}'^\top$.
Since $\text{colsp}\,[\boldsymbol{X}'] = \text{colsp}\,[\boldsymbol{X}]$, both H and H' project onto the same subspace.
Orthogonal projection matrices onto a given subspace are unique.
Therefore $H = H'$.

(e) [easy] Generalize the previous problem by proving that orthogonal projection matrices that project onto any specific subspace are *unique*.

Let V be a subspace.
H_1 and H_2 are both orthogonal projection matrices onto V.
Then for any vector y, both $H_1 y$ and $H_2 y$ are the unique vector in V closest to y.
So $H_1 y = H_2 y$ for every y.
Therefore $H_1 = H_2$.
So orthogonal projection matrices onto a given subspace are unique.

(f) [easy] Prove that if a square matrix is both symmetric and idempotent then it must be an orthogonal projection matrix.

Symmetric means $H^\top = H$ and idempotent means $H^2 = H$.
For any vector y, let $\hat{y} = Hy$.
Then $\hat{y} \in colsp(H)$.
Also $y - \hat{y} = y - Hy$.
Now $H(y - \hat{y}) = H(y - Hy) = Hy - H^2 y = Hy - Hy = 0$.
So $y - \hat{y} \in null(H)$.
Since H is symmetric, $null(H)$ is orthogonal to $colsp(H)$.
Thus $y - \hat{y}$ is orthogonal to colsp $[(]\, H)$, so Hy is the orthogonal projection of y onto colsp $[(]\, H)$.
Therefore H is an orthogonal projection matrix.

(g) [difficult] [MA] Prove the converse of the previous question: that if a square matrix is an orthogonal projection matrix, then it must be both symmetric and idempotent.

Let H be an orthogonal projection matrix onto a subspace V.
For any vector y, write $y = v + w$ where $v \in V$ and $w \perp V$.
Since H projects onto V, we have $Hy = v$.
Now apply H again: $H^2 y = H(Hy) = H(v) = v = Hy$.
So $H^2 y = Hy$ for every y.
Therefore $H^2 = H$, so H is idempotent.
Now let $y_1 = v_1 + w_1$ and $y_2 = v_2 + w_2$ where $v_1, v_2 \in V$ and $w_1, w_2 \perp V$.
Then $Hy_1 = v_1$ and $Hy_2 = v_2$.
So $\langle Hy_1, y_2 \rangle = \langle v_1, v_2 + w_2 \rangle = \langle v_1, v_2 \rangle$ since $v_1 \perp w_2$.
Also $\langle y_1, Hy_2 \rangle = \langle v_1 + w_1, v_2 \rangle = \langle v_1, v_2 \rangle$ since $w_1 \perp v_2$.
So $\langle Hy_1, y_2 \rangle = \langle y_1, Hy_2 \rangle$ for all y_1, y_2.
So $H^\top = H$, so H is symmetric.
Hence if a square matrix is an orthogonal projection matrix, then it is both symmetric and idempotent.
(used ChatGPT)

(h) [easy] Prove that I_n is an orthogonal projection matrix $\forall n$.

$\boldsymbol{I}_n^\top = \boldsymbol{I}_n$, so $\boldsymbol{I}_n$ is symmetric.
Also $\boldsymbol{I}_n^2 = \boldsymbol{I}_n$, so $\boldsymbol{I}_n$ is idempotent.
Therefore $\boldsymbol{I}_n$ is an orthogonal projection matrix.

(i) [easy] What subspace does I_n project onto?

$\boldsymbol{I}_n y = y$ for every $y \in \mathbb{R}^n$.
So $\boldsymbol{I}_n$ projects every vector onto itself.
Therefore $\boldsymbol{I}_n$ projects onto $\mathbb{R}^n$.

(j) [easy] Consider least squares linear regression using a design matrix X with rank $p+1$. What are the degrees of freedom in the resulting model? What does this mean?

If $\textbf{rank}(\boldsymbol{X}) = p + 1$, then the degrees of freedom are $p + 1$.
This means $colsp(X)$ has dimension $p + 1$.
So the fitted vector $\hat{y} = Hy$ lies in a $(p + 1)$-dimensional subspace, and we are estimating $p + 1$ coefficients: $b_0, b_1, \ldots, b_p$.

(k) [difficult] We saw that the perceptron is an *iterative algorithm*. This means that it goes through multiple iterations in order to converge to a closer and closer $\boldsymbol{w}$. Why not do the same with linear least squares regression? Consider the following. Regress $\boldsymbol{y}$ using $\boldsymbol{X}$ to get $\hat{\boldsymbol{y}}$. This generates residuals $\boldsymbol{e}$ (the leftover piece of $\boldsymbol{y}$ that wasn't explained by the regression's fit, $\hat{\boldsymbol{y}}$). Now try again! Regress $\boldsymbol{e}$ using $\boldsymbol{X}$ and then get new residuals $\boldsymbol{e}_{new}$. Would $\boldsymbol{e}_{new}$ be closer to $\boldsymbol{0}_n$ than the first $\boldsymbol{e}$? That is, wouldn't this yield a better model on iteration #2? Yes/no and explain.

No.
After regressing y on $\boldsymbol{X}$, we get $\hat{y} = Hy$ and residuals $e = y - \hat{y}$.
From OLS, e is orthogonal to colsp $[\boldsymbol{X}]$.
So if we regress e on $\boldsymbol{X}$, its projection onto colsp $[\boldsymbol{X}]$ is 0.
Thus the new fitted values are 0_n and the new residuals are $e_{new} = e$.
Therefore iteration #2 does not improve the model.
(help from ChatGPT)

(l) [harder] Prove that $\boldsymbol{Q}^\top = \boldsymbol{Q}^{-1}$ where $\boldsymbol{Q}$ is an orthonormal matrix such that colsp $[\boldsymbol{Q}] =$ colsp $[\boldsymbol{X}]$ and $\boldsymbol{Q}$ and $\boldsymbol{X}$ are both matrices $\in \mathbb{R}^{n \times (p+1)}$ and $n = p + 1$ in this case to ensure the inverse is defined. Hint: this is purely a linear algebra exercise and it's a one-liner.

Since $\boldsymbol{Q}$ is orthonormal, $\boldsymbol{Q}^\top \boldsymbol{Q} = \boldsymbol{I}_n$.
Therefore $\boldsymbol{Q}^\top = \boldsymbol{Q}^{-1}$

(m) [easy] Prove that the least squares projection $H = \boldsymbol{X}\left(\boldsymbol{X}^T\boldsymbol{X}\right)^{-1}\boldsymbol{X}^T = \boldsymbol{Q}\boldsymbol{Q}^\top$. Justify each step.

Since $\text{colsp}[\boldsymbol{Q}] = \text{colsp}[\boldsymbol{X}]$, both $\boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top$ and $\boldsymbol{Q}\boldsymbol{Q}^\top$ are orthogonal projection matrices onto the same subspace.
Orthogonal projection matrices onto a given subspace are unique.
Therefore $\boldsymbol{X}(\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top = \boldsymbol{Q}\boldsymbol{Q}^\top$.

(n) [difficult] [MA] This problem is independent of the others. Let H be an orthogonal projection matrix. Prove that $\text{rank}(H) = \text{tr}(H)$. Hint: you will need to use facts about eigenvalues and the eigendecomposition of projection matrices.

Let H be an orthogonal projection matrix.
Then H is symmetric and idempotent, so $H^2 = H$.
Let λ be an eigenvalue of H with eigenvector v, so $Hv = \lambda v$.
Apply H again:
$H^2v = H(Hv) = H(\lambda v) = \lambda Hv = \lambda^2 v$.
But $H^2 = H$, so $H^2v = Hv = \lambda v$.
So $\lambda^2 v = \lambda v$, so $\lambda^2 = \lambda$.
So $\lambda(\lambda - 1) = 0$, which means the eigenvalues are 0 or 1.
$\text{tr}[(]\, H)$ is the sum of the eigenvalues, so it equals the number of 1's.
$\text{rank}[(]\, H)$ is the number of nonzero eigenvalues, which is also the number of 1's.
Therefore $\text{rank}[(]\, H) = \text{tr}[(]\, H)$.

(o) [harder] Prove that an orthogonal projection onto the $\text{colsp}[\boldsymbol{Q}]$ is the same as the sum of the projections onto each column of $\boldsymbol{Q}$.

Let $\boldsymbol{Q} = [q_1, \ldots, q_k]$ where the columns are orthonormal.
The orthogonal projection onto colsp $[\boldsymbol{Q}]$ is $\boldsymbol{Q}\boldsymbol{Q}^\top$.
Write $\boldsymbol{Q}\boldsymbol{Q}^\top = [q_1, \ldots, q_k] \begin{bmatrix} q_1^\top \\ \vdots \\ q_k^\top \end{bmatrix}$.
So $\boldsymbol{Q}\boldsymbol{Q}^\top = \sum_{j=1}^{k} q_j q_j^\top$.
But $q_j q_j^\top y$ is the projection of y onto the line spanned by q_j.
Therefore the projection onto colsp $[\boldsymbol{Q}]$ is the sum of the projections onto each column of $\boldsymbol{Q}$.
(used ChatGPT for formatting)

(p) [easy] Explain why adding a new column to $\boldsymbol{X}$ results in no change to SST.

$SST = \sum_{i=1}^{n}(y_i - \bar{y})^2$ depends only on y.
It does not depend on $\boldsymbol{X}$.
So adding a new column to $\boldsymbol{X}$ does not change SST.

(q) [harder] Prove that adding a new column to $\boldsymbol{X}$ results in SSR increasing.

Adding a new column to $\boldsymbol{X}$ makes colsp $[\boldsymbol{X}]$ larger or keeps it the same.
So the projection $\hat{y}$ is onto a larger subspace.
A projection onto a larger subspace cannot reduce the squared length of the projection.
Since $SSR = ||\hat{y} - \bar{y}1_n||^2$, SSR increases or stays the same.

(r) [harder] What is overfitting? Use what you learned in this problem to frame your answer.

Overfitting occurs when we add more columns to $\boldsymbol{X}$ and improve the in-sample fit but only by fitting noise instead of signal.
In this problem, adding columns makes SSR increase and SSE decrease in-sample.
But this does not mean the model predicts new data better.
So overfitting is making the model look better on the training data without improving true prediction.

(s) [easy] Why are the "in-sample" error metrics R^2 and SSE dishonest? Note: I'm leaving out MSE and RMSE as they attempt to be honest by increasing as p increases due to the denominator.

In-sample R^2 and SSE are dishonest because they are computed on the same data used to fit the model.
As we add more columns to $\boldsymbol{X}$, the projection subspace gets larger.
So SSE can only go down and R^2 can only go up even if the new variables are just noise.

(t) [easy] How can we provide honest error metrics for R^2, SSE? It may help to draw a picture of the procedure.

To get honest error metrics, split the data into a training set and a test set.
Fit the model using the training set.
Then compute SSE and R^2 using predictions on the test set.

(u) [easy] The procedure in the previous question produces highly variable honest error metrics. Can you change the procedure slightly to reduce the variation in the honest error metrics? What is this procedure called and how is it done?

Yes.
Instead of one train/test split, repeat the process multiple times and average the test errors.
This is called cross-validation.
In K-fold cross-validation, divide the data into K parts, train on $K-1$ parts, test on the remaining part, and repeat until each part has been the test set once.
Then average the error metrics.

Problem 4

These are some questions related to validation.

(a) [easy] Assume you are doing one train-test split where you build the model on the training set and validate on the test set. What does the constant K control? And what is its tradeoff?

In a train–test split, K controls the proportion of the data used for training versus testing.
A larger training set gives a better fitted model but leaves less data for testing.
A larger test set gives a more reliable estimate of prediction error but leaves less data to train the model.
So the tradeoff is between model accuracy and reliability of the error estimate.

(b) [easy] What problem does K-fold CV try to solve?

K-fold cross-validation reduces the randomness from using only one train–test split. Different splits can give different error estimates. K-fold CV averages multiple splits to get a more stable estimate.

(c) [difficult] [MA] Theoretically, how does K-fold CV solve this problem? The Internet is your friend.

In K-fold cross-validation, the data is divided into K parts.
Each time, the model is trained on $K-1$ parts and tested on the remaining part.
This is repeated K times so every part is used once as the test set.
The test errors are then averaged to estimate the true prediction error.